



For: Anoop Kumar

Position: Senior Manager, Programs

Organization: Walmart

To Whom It May Concern,

It is my privilege to write this strong recommendation in support of Anoop Kumar's EB1A application for extraordinary ability. I am Rahul Vishwakarma, an IEEE Senior Member, researcher, and inventor with over 60 granted U.S. patents, and extensive experience in AI, ML, and secure cloud computing systems. Currently, I am Senior Machine Learning Engineer at WorkOnward Inc and I have served as an Entrepreneur-in-Residence at Robin Hood Blue Ridge Labs and also an Ambassador with the International Society of Service Innovation Professionals (ISSIP). Having reviewed over 40 technical papers and mentored emerging AI professionals, I can attest that Anoop stands in the topmost tier of talent globally.

I had the honor of working alongside Anoop as a Grand Awards Judge at the International Science and Engineering Fair (ISEF) 2024—one of the world's most competitive research showcases for high school students. Judges are selected for their distinguished academic and industry credentials. Anoop's appointment reflects his deep technical expertise, international recognition, and impact in AI and software innovation. His sharp analytical thinking, fairness, and depth in evaluating novel software architectures exemplified why he is widely considered a thought leader in the field.

Transformative Impact at Walmart: Leading AI-Driven Programs

Anoop serves as Senior Technical Program Manager at Walmart, where he has led AI-first digital transformation initiatives across their \$100B+ e-commerce division. Two landmark programs stand out:

- **Conversational AI Implementation:** He pioneered the rollout of AI-powered shopping assistants and voice-based product discovery. This AI integration not only improved customer experience but also streamlined backend service workflows, enhancing call center resolution rates by 42%.
- **Generative AI for Product Catalog Optimization:** Anoop led Walmart's use of Generative AI to enrich over 850 million product listings. This resulted in a 37% boost in personalized recommendations, improved search accuracy, and elevated conversion rates—yielding \$456 million in incremental GMV.

In supply chain operations, Anoop drove predictive AI models that improved Walmart's inventory forecasting accuracy by 28%, reducing overstock and stockout scenarios. His



programs achieved an estimated \$380 million in cost optimizations, significantly enhancing Walmart's operational agility in a highly competitive retail landscape.

Academic Contributions and Thought Leadership

Anoop's research portfolio includes 14 peer-reviewed papers across IEEE, Springer, and international conferences. His work is not only academically rigorous but also industrially relevant:

- In *"Exploring Strategies for Privacy-Preserving Machine Learning"*, he introduces secure federated learning techniques essential to safeguarding sensitive consumer data in AI systems.
- In *"AI-Driven Self-Healing Cloud Systems"*, he proposes novel anomaly detection frameworks for automated recovery in enterprise-grade cloud environments.
- His IEEE paper on *"Enhancing Security in Cloud-Native Applications"* advances the discourse on AI-led cybersecurity, offering scalable blueprints for mitigating evolving threats in zero-trust architectures.

These contributions strengthen the bridge between theoretical AI and real-world enterprise deployments—essential to both academia and industry.

Technological Innovation: Patents and Applied Impact

Anoop's six UK-registered design patents demonstrate his exceptional ability to turn abstract AI concepts into scalable, real-world solutions:

- His **AI-Based Energy Monitoring Device** has achieved up to 25% reduction in electricity usage—already piloted in Walmart distribution centers for sustainability optimization.
- His **Data Visualization Device for ML Analytics** improves dashboard rendering by 50%, enabling faster executive decision-making across large datasets.
- His **ML-Based Collision Avoidance System for Drones** reduces aerial incident rates, supporting logistics and military applications.
- His patented **AI-Integrated CRM Interfaces** have shown a 35% uplift in lead conversion and 20% drop in churn, powering next-gen customer engagement platforms.

Each patent targets a high-impact vertical—retail, defense, sustainability, and enterprise software—making Anoop's portfolio not only technically rich but commercially vital.

Recognition and Competitive Honors



Anoop has been honored as a keynote speaker at TechEX and Intelligent Automation North America, where he presented on AI-driven retail transformation. He has judged elite awards such as the Stevie and Webby Awards—competitions that rigorously vet entrants across global domains of business excellence and digital innovation. These recognitions are reserved for professionals who exemplify domain mastery, cross-industry influence, and measurable impact.

Conclusion

Anoop Kumar's contributions to AI research, enterprise-scale digital transformation, and patented innovation clearly distinguish him as a professional of extraordinary ability. His leadership in deploying AI for national-scale operations, his scholarly research, and the international adoption of his patented technologies place him among the top echelon of AI leaders.

I offer my strongest endorsement for his EB1A petition and am happy to provide further support if needed.

Sincerely,

Signed by:

A handwritten signature in black ink, appearing to read 'Rahul Deo Vishwakarma', enclosed within a blue digital signature box.

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Rahul Deo Vishwakarma

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